Causal Freshness-Aware Secure HAPS–ISAC System Model

Signal Model, Target Tracking, Information Ages, and Optimisation Problem

Working Technical Model

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Abstract

This document specifies a complete, algorithm-independent system and signal model for a secure non-terrestrial integrated sensing and communication (ISAC) network. A fixed full-duplex high-altitude platform station (HAPS) serves geographically distributed ground Internet-of-Things (IoT) devices arranged in near–far non-orthogonal multiple-access (NOMA) pairs, illuminates and tracks one mobile ground target that may eavesdrop on the IoT traffic, and coordinates a friendly aerial jamming vehicle (AAV). A quasi-stationary aerial reconfigurable intelligent surface (RIS) assists signal steering. The model explicitly includes direct and RIS-assisted channels, imperfect channel state information (CSI), residual self-interference, imperfect successive interference cancellation (SIC), sensing and processing delay, target-state uncertainty, discrete RIS phases, control overhead, and AAV motion and energy. Legitimate update freshness, leaked-information staleness, and target-estimate freshness are represented by age of information (AoI), age of leaked information (AoLI), and age of sensing information (AoSI), respectively. A causal long-term stochastic optimisation problem is then formulated without committing to a particular solution algorithm.

1 Modelling scope and slot chronology

The baseline is deliberately restricted to one HAPS, one fixed or slowly hovering aerial RIS, one friendly AAV, one mobile ground target/eavesdropper, and fixed NOMA user pairing. MEC offloading, multiple targets, mobile RIS placement, and dynamic pairing are progressive extensions rather than baseline assumptions. This separation keeps the core problem scientifically identifiable and computationally workable.

Time is slotted as $t \in \mathcal{T} \triangleq \{0, 1, \ldots\}$, with slot duration T_s. Channels and control variables are constant within a slot. The causal order is:

1. at the start of slot t, the HAPS observes current channel estimates, information ages, the latest *available* target belief, and the AAV state;
2. it schedules at most one NOMA pair and selects its communication beam, NOMA powers, sensing beam, RIS phases, receiver combiner, HAPS CPU frequency, and the AAV command;
3. the HAPS transmits the joint communication–sensing waveform while receiving the target echo, and the AAV transmits artificial noise;
4. legitimate and intercepted packet outcomes update AoI and AoLI;
5. the echo is detected and processed on board the HAPS; after a non-zero processing delay, the accepted target estimate updates AoSI and target uncertainty;
6. this new estimate may affect AAV and beam decisions only in a later slot, never retroactively in slot t.

This chronology removes the common but physically invalid assumption that a same-slot radar echo can steer the same-slot AAV action.

2 Network geometry and entities

The fixed HAPS position is $\boldsymbol{q}_\mathrm{H} = [x_\mathrm{H}, y_\mathrm{H}, H_\mathrm{H}]^\mathrm{T}$, with N_t transmit antennas and N_r receive antennas. There are $K = 2M$ single-antenna IoT devices partitioned into M fixed near–far pairs

$$\mathcal{U}_m = \{(m, \mathrm{n}), (m, \mathrm{f})\}, \qquad m \in \mathcal{M} \triangleq \{1, \ldots, M\}, \tag{1}$$

at known positions $\boldsymbol{q}_{m,k} = [x_{m,k}, y_{m,k}, 0]^\mathrm{T}$, $k \in \{\mathrm{n}, \mathrm{f}\}$. "Near" and "far" refer to the long-term effective channel order used to establish a stable SIC order; the order can be recomputed on a slower network-management timescale.

The aerial RIS has L passive elements and a fixed/hovering position $\boldsymbol{q}_\mathrm{R}$. The mobile target, which is also the potential eavesdropper, has position $\boldsymbol{q}_\mathrm{T}[t] = [x_\mathrm{T}[t], y_\mathrm{T}[t], 0]^\mathrm{T}$. The friendly AAV flies at altitude H_J with position $\boldsymbol{q}_\mathrm{J}[t] = [x_\mathrm{J}[t], y_\mathrm{J}[t], H_\mathrm{J}]^\mathrm{T}$ and carries N_J transmit antennas.

Pair scheduling is represented by

$$a_m[t] \in \{0, 1\}, \qquad \sum_{m=1}^{M} a_m[t] \le 1. \tag{2}$$

Thus, the two devices in the selected pair share the same time–frequency resource by power-domain NOMA, while different pairs are orthogonal in time. This avoids unmodelled inter-pair interference and remains compatible with the wide HAPS coverage area.

3 Channel and RIS model

3.1 Large- and small-scale fading

For an aerial/terrestrial link from node a to node b, a general Rician block-fading model is

$$\boldsymbol{H}_{ab}[t] = \sqrt{\beta_{ab}[t]} \left(\sqrt{\frac{\kappa_{ab}}{1+\kappa_{ab}}}\, \overline{\boldsymbol{H}}_{ab}[t] + \sqrt{\frac{1}{1+\kappa_{ab}}}\, \widetilde{\boldsymbol{H}}_{ab}[t] \right), \tag{3}$$

where κ_{ab} is the Rician factor, $\overline{\boldsymbol{H}}_{ab}$ is the array-response LoS component, and the entries of $\widetilde{\boldsymbol{H}}_{ab}$ are zero-mean unit-variance complex Gaussian variables. A suitable elevation-aware average power gain is

$$\beta_{ab}[t] = 10^{-\mathrm{PL}_{ab}[t]/10}, \tag{4}$$

$$\mathrm{PL}_{ab}[t] = \mathrm{FSPL}(d_{ab}[t]) + P_{ab}^{\mathrm{LoS}}[t]\eta_{\mathrm{LoS}} + (1 - P_{ab}^{\mathrm{LoS}}[t])\eta_{\mathrm{NLoS}} + A_{ab}[t], \tag{5}$$

where $d_{ab} = \|\boldsymbol{q}_a - \boldsymbol{q}_b\|$, P_{ab}^{LoS} is an environment- and elevation-dependent LoS probability, A_{ab} represents atmospheric/rain attenuation when relevant, and $\mathrm{FSPL}(d) = 20\log_{10}(4\pi f_c d/c)$. Equation (5) can be replaced by a measurement-calibrated NTN path-loss law without changing the subsequent formulation.

3.2 Aerial RIS

The RIS coefficient matrix is

$$\boldsymbol{\Theta}[t] = \mathrm{diag}\Big(\eta_1 e^{j\theta_1[t]}, \ldots, \eta_L e^{j\theta_L[t]}\Big), \quad 0 < \eta_\ell \le 1, \tag{6}$$

with b_θ-bit phase shifts

$$\theta_\ell[t] \in \mathcal{F}_{b_\theta} \triangleq \left\{ \frac{2\pi r}{2^{b_\theta}} : r = 0, \ldots, 2^{b_\theta} - 1 \right\}. \tag{7}$$

Let $\boldsymbol{G}_{\mathrm{HR}}[t] \in \mathbb{C}^{L\times N_{\mathrm{t}}}$ denote the HAPS–RIS channel, $\boldsymbol{h}_{\mathrm{Hv}}[t] \in \mathbb{C}^{N_{\mathrm{t}}}$ the direct HAPS–node channel, and $\boldsymbol{h}_{\mathrm{Rv}}[t] \in \mathbb{C}^{L}$ the RIS–node channel. For $v \in \{(m,k), \mathrm{T}\}$, the effective downlink channel is

$$\boldsymbol{g}_v^{\mathrm{H}}[t] = \boldsymbol{h}_{\mathrm{Hv}}^{\mathrm{H}}[t] + \boldsymbol{h}_{\mathrm{Rv}}^{\mathrm{H}}[t]\boldsymbol{\Theta}[t]\boldsymbol{G}_{\mathrm{HR}}[t]. \tag{8}$$

The controller has imperfect CSI

$$\boldsymbol{g}_v[t] = \widehat{\boldsymbol{g}}_v[t] + \boldsymbol{e}_v[t], \qquad \boldsymbol{e}_v[t] \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{C}_v[t]), \tag{9}$$

or, equivalently, a bounded uncertainty set $\|\boldsymbol{e}_v\| \le \varepsilon_v$. The statistical and bounded models lead to chance-constrained and worst-case robust variants, respectively.

RIS control consumes time. One explicit switching-overhead model is

$$\tau_{\mathrm{RIS}}[t] = \tau_0 \sum_{\ell=1}^{L} \mathbf{1}\{\theta_\ell[t] \ne \theta_\ell[t-1]\}, \quad T_{\mathrm{d}}[t] = [T_{\mathrm{s}} - \tau_{\mathrm{fix}} - \tau_{\mathrm{RIS}}[t]]^+, \tag{10}$$

where T_{d} is the payload-bearing duration. A practical constraint imposes $T_{\mathrm{d}}[t] \ge T_{\min} > 0$. If the RIS is controlled only every T_{R} slots, then $\boldsymbol{\Theta}[t] = \boldsymbol{\Theta}[t-1]$ for $t \not\equiv 0 \pmod{T_{\mathrm{R}}}$.

4 HAPS and AAV transmitted signals

Let $s_{m,k}[t]$ be an independent unit-power update symbol for device (m,k) and $s_{\mathrm{s}}[t]$ a unit-power sensing sequence known to the HAPS receiver. The HAPS signal is

$$\boldsymbol{x}_{\mathrm{H}}[t] = \sum_{m=1}^{M} a_m[t]\boldsymbol{v}_m[t] \left(\sqrt{p_{m,\mathrm{n}}[t]}s_{m,\mathrm{n}}[t] + \sqrt{p_{m,\mathrm{f}}[t]}s_{m,\mathrm{f}}[t] \right) + \boldsymbol{w}_{\mathrm{s}}[t]s_{\mathrm{s}}[t], \tag{11}$$

where $\|\boldsymbol{v}_m[t]\| = 1$ is a pair beam, $p_{m,k}[t] \ge 0$ are NOMA powers, and $\boldsymbol{w}_{\mathrm{s}}[t]$ is the sensing beam. Its covariance is

$$\boldsymbol{Q}_{\mathrm{H}}[t] = \mathbb{E}\{\boldsymbol{x}_{\mathrm{H}}\boldsymbol{x}_{\mathrm{H}}^{\mathrm{H}}\} = \sum_{m} a_m(p_{m,\mathrm{n}} + p_{m,\mathrm{f}})\boldsymbol{v}_m\boldsymbol{v}_m^{\mathrm{H}} + \boldsymbol{w}_{\mathrm{s}}\boldsymbol{w}_{\mathrm{s}}^{\mathrm{H}}, \tag{12}$$

subject to $\mathrm{tr}(\boldsymbol{Q}_{\mathrm{H}}[t]) \leq P_{\mathrm{H}}^{\max}$. The far user is normally assigned at least as much power as the near user,

$$p_{m,\mathrm{f}}[t] \geq p_{m,\mathrm{n}}[t] \geq 0. \tag{13}$$

The AAV sends independent artificial noise $z_{\mathrm{J}}[t] \sim \mathcal{CN}(0,1)$:

$$\boldsymbol{x}_{\mathrm{J}}[t] = \sqrt{p_{\mathrm{J}}[t]}\boldsymbol{u}_{\mathrm{J}}[t]z_{\mathrm{J}}[t], \quad 0 \leq p_{\mathrm{J}}[t] \leq P_{\mathrm{J}}^{\max}, \quad \|\boldsymbol{u}_{\mathrm{J}}[t]\| = 1. \tag{14}$$

The jamming beam is designed to concentrate energy toward the *predicted* target location while limiting interference at scheduled IoT devices.

5 Downlink communication and NOMA decoding

Let $\boldsymbol{f}_{\mathrm{J},v}[t] \in \mathbb{C}^{N_{\mathrm{J}}}$ be the AAV–node channel. A scheduled device receives

$$y_{m,k}[t] = \boldsymbol{g}_{m,k}^{\mathrm{H}}[t]\boldsymbol{x}_{\mathrm{H}}[t] + \sqrt{p_{\mathrm{J}}[t]}\boldsymbol{f}_{\mathrm{J},m,k}^{\mathrm{H}}[t]\boldsymbol{u}_{\mathrm{J}}[t]z_{\mathrm{J}}[t] + n_{m,k}[t], \tag{15}$$

where $n_{m,k} \sim \mathcal{CN}(0, \sigma_{m,k}^2)$. Define

$$G_{m,k}[t] = \left|\boldsymbol{g}_{m,k}^{\mathrm{H}}[t]\boldsymbol{v}_m[t]\right|^2, \tag{16}$$

$$I_{m,k}^{\mathrm{s}}[t] = \xi_{m,k}^{\mathrm{s}}\left|\boldsymbol{g}_{m,k}^{\mathrm{H}}[t]\boldsymbol{w}_{\mathrm{s}}[t]\right|^2, \tag{17}$$

$$I_{m,k}^{\mathrm{J}}[t] = p_{\mathrm{J}}[t]\left|\boldsymbol{f}_{\mathrm{J},m,k}^{\mathrm{H}}[t]\boldsymbol{u}_{\mathrm{J}}[t]\right|^2, \tag{18}$$

$$D_{m,k}[t] = I_{m,k}^{\mathrm{s}}[t] + I_{m,k}^{\mathrm{J}}[t] + \sigma_{m,k}^2, \tag{19}$$

where $0 \leq \xi_{m,k}^{\mathrm{s}} \leq 1$ captures cancellation of the known sensing waveform at an IoT receiver. For the selected pair, the far user decodes its own message while treating the near message as interference:

$$\gamma_{m,\mathrm{f}}[t] = \frac{p_{m,\mathrm{f}}G_{m,\mathrm{f}}}{p_{m,\mathrm{n}}G_{m,\mathrm{f}} + D_{m,\mathrm{f}}}. \tag{20}$$

The near user first decodes the far message and then its own message:

$$\gamma_{m,\mathrm{n}\to\mathrm{f}}[t] = \frac{p_{m,\mathrm{f}}G_{m,\mathrm{n}}}{p_{m,\mathrm{n}}G_{m,\mathrm{n}} + D_{m,\mathrm{n}}}, \tag{21}$$

$$\gamma_{m,\mathrm{n}}[t] = \frac{p_{m,\mathrm{n}}G_{m,\mathrm{n}}}{\epsilon_{\mathrm{SIC}}p_{m,\mathrm{f}}G_{m,\mathrm{n}} + D_{m,\mathrm{n}}}, \tag{22}$$

where $0 \leq \epsilon_{\mathrm{SIC}} \leq 1$ is the residual fraction after imperfect SIC. The effective rates (bit/s) are

$$R_{m,\mathrm{f}}[t] = W\log_2(1 + min\{\gamma_{m,\mathrm{f}}, \gamma_{m,\mathrm{n}\to\mathrm{f}}\}), \tag{23}$$

$$R_{m,\mathrm{n}}[t] = W\log_2(1 + \gamma_{m,\mathrm{n}}), \tag{24}$$

where W is bandwidth. For a fresh payload of $L_{m,k}$ bits generated at the start of slot t, successful delivery is

$$\chi_{m,k}[t] = a_m[t]\mathbf{1}\{T_{\mathrm{d}}[t]R_{m,k}[t] \geq L_{m,k}\}. \tag{25}$$

Equation (25) includes the RIS/control overhead and therefore does not equate a high instantaneous SINR with a completed update.

5.1 Reception at the target/eavesdropper

The mobile target receives

$$y_{\mathrm{T}}[t] = \boldsymbol{g}_{\mathrm{T}}^{\mathrm{H}}[t]\boldsymbol{x}_{\mathrm{H}}[t] + \sqrt{p_{\mathrm{J}}[t]}\boldsymbol{f}_{\mathrm{J},\mathrm{T}}^{\mathrm{H}}[t]\boldsymbol{u}_{\mathrm{J}}[t]z_{\mathrm{J}}[t] + n_{\mathrm{T}}[t]. \tag{26}$$

To avoid understating leakage, the target is conservatively allowed to attempt the same SIC order. Define

$$G_{\mathrm{T}}[t] = \left|\boldsymbol{g}_{\mathrm{T}}^{\mathrm{H}}[t]\boldsymbol{v}_m[t]\right|^2, \tag{27}$$

$$D_{\mathrm{T}}[t] = \xi_{\mathrm{T}}^{\mathrm{s}}\left|\boldsymbol{g}_{\mathrm{T}}^{\mathrm{H}}[t]\boldsymbol{w}_{\mathrm{s}}[t]\right|^2 + p_{\mathrm{J}}[t]\left|\boldsymbol{f}_{\mathrm{J},\mathrm{T}}^{\mathrm{H}}[t]\boldsymbol{u}_{\mathrm{J}}[t]\right|^2 + \sigma_{\mathrm{T}}^2, \tag{28}$$

$$\gamma_{\mathrm{T},m,\mathrm{f}}[t] = \frac{p_{m,\mathrm{f}}G_{\mathrm{T}}}{p_{m,\mathrm{n}}G_{\mathrm{T}} + D_{\mathrm{T}}}, \tag{29}$$

$$\gamma_{\mathrm{T},m,\mathrm{n}}[t] = \frac{p_{m,\mathrm{n}}G_{\mathrm{T}}}{\epsilon_{\mathrm{T}}p_{m,\mathrm{f}}G_{\mathrm{T}} + D_{\mathrm{T}}}, \tag{30}$$

and $R_{\mathrm{T},m,k}[t] = W \log_2(1 + \gamma_{\mathrm{T},m,k}[t])$. If the target cannot perform SIC, set $\epsilon_{\mathrm{T}} = 1$ for the near stream. The interception event and instantaneous secrecy rate are

$$\ell_{m,k}[t] = a_m[t]\mathbf{1}\{T_{\mathrm{d}}[t]R_{\mathrm{T},m,k}[t] \geq L_{m,k}\}\,, \tag{31}$$

$$R^{\mathrm{sec}}_{m,k}[t] = [R_{m,k}[t] - R_{\mathrm{T},m,k}[t]]^{+}\,. \tag{32}$$

The model distinguishes a rate-based secrecy outage from an actual packet interception; both are useful because AoLI changes only when a complete fresh payload is intercepted.

6 Full-duplex sensing and echo processing

6.1 Echo signal and sensing SINR

The HAPS operates in full duplex: it receives the echo while transmitting $\boldsymbol{x}_{\mathrm{H}}[t]$. A compact target response is

$$\boldsymbol{H}^{\mathrm{e}}_{\mathrm{T}}[t;\boldsymbol{\Theta}] = \alpha_{\mathrm{T}}[t;\boldsymbol{\Theta}]\,\boldsymbol{a}_{\mathrm{r}}(\vartheta_{\mathrm{T}}[t])\boldsymbol{a}^{\mathrm{H}}_{\mathrm{t}}(\vartheta_{\mathrm{T}}[t]), \tag{33}$$

where α_{T} contains the two-way propagation loss, radar cross section, phase, and any RIS-assisted illumination component. This form permits a direct-only echo, an RIS-assisted forward path, or a calibrated composite echo model. The received array signal is

$$\boldsymbol{y}_{\mathrm{s}}[t] = \boldsymbol{H}^{\mathrm{e}}_{\mathrm{T}}[t]\boldsymbol{x}_{\mathrm{H}}[t] + \sqrt{\rho_{\mathrm{SI}}}\boldsymbol{H}_{\mathrm{SI}}[t]\boldsymbol{x}_{\mathrm{H}}[t] + \boldsymbol{H}_{\mathrm{JH}}[t]\boldsymbol{x}_{\mathrm{J}}[t] + \boldsymbol{n}_{\mathrm{s}}[t], \tag{34}$$

where $0 \leq \rho_{\mathrm{SI}} \ll 1$ is the residual self-interference (SI) power factor after analogue/digital cancellation, $\boldsymbol{H}_{\mathrm{JH}}$ is the AAV–HAPS channel, and $\boldsymbol{n}_{\mathrm{s}} \sim \mathcal{CN}(\mathbf{0}, \sigma^2_{\mathrm{s}}\boldsymbol{I})$. Because the HAPS knows its own communication data and sensing sequence, the whole dual-functional waveform can contribute to sensing after matched processing. For receive combiner $\|\boldsymbol{u}_{\mathrm{s}}[t]\| = 1$, an effective sensing SINR is

$$\gamma_{\mathrm{s}}[t] = \frac{\boldsymbol{u}^{\mathrm{H}}_{\mathrm{s}}\boldsymbol{H}^{\mathrm{e}}_{\mathrm{T}}\boldsymbol{Q}_{\mathrm{H}}(\boldsymbol{H}^{\mathrm{e}}_{\mathrm{T}})^{\mathrm{H}}\boldsymbol{u}_{\mathrm{s}}}{\boldsymbol{u}^{\mathrm{H}}_{\mathrm{s}}\left(\rho_{\mathrm{SI}}\boldsymbol{H}_{\mathrm{SI}}\boldsymbol{Q}_{\mathrm{H}}\boldsymbol{H}^{\mathrm{H}}_{\mathrm{SI}} + p_{\mathrm{J}}\boldsymbol{H}_{\mathrm{JH}}\boldsymbol{u}_{\mathrm{J}}\boldsymbol{u}^{\mathrm{H}}_{\mathrm{J}}\boldsymbol{H}^{\mathrm{H}}_{\mathrm{JH}} + \sigma^2_{\mathrm{s}}\boldsymbol{I}\right)\boldsymbol{u}_{\mathrm{s}}}. \tag{35}$$

The detection indicator is $d_{\mathrm{s}}[t] = \mathbf{1}\{\gamma_{\mathrm{s}}[t] \geq \Gamma_{\mathrm{s}}\}$. A calibrated detection law $P_{\mathrm{D}} = f_{\mathrm{D}}(\gamma_{\mathrm{s}}, P_{\mathrm{FA}})$ can replace this threshold abstraction.

6.2 Target dynamics, measurement, and belief update

For a nearly constant-velocity ground target,

$$\boldsymbol{x}_{\mathrm{T}}[t] = [x_{\mathrm{T}}, y_{\mathrm{T}}, v_{\mathrm{T},x}, v_{\mathrm{T},y}]^{\mathrm{T}}, \tag{36}$$

$$\boldsymbol{x}_{\mathrm{T}}[t+1] = \boldsymbol{F}\boldsymbol{x}_{\mathrm{T}}[t] + \boldsymbol{w}_{\mathrm{T}}[t], \qquad \boldsymbol{w}_{\mathrm{T}}[t] \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q}_{\mathrm{T}}), \tag{37}$$

$$\boldsymbol{F} = \begin{bmatrix} 1 & 0 & T_{\mathrm{s}} & 0 \\ 0 & 1 & 0 & T_{\mathrm{s}} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{38}$$

When $d_{\mathrm{s}}[t] = 1$, the processed measurement may contain range, angle of arrival (AoA), and Doppler,

$$\boldsymbol{z}_{\mathrm{s}}[t] = \boldsymbol{h}(\boldsymbol{x}_{\mathrm{T}}[t]) + \boldsymbol{v}_{\mathrm{s}}[t], \qquad \boldsymbol{v}_{\mathrm{s}}[t] \sim \mathcal{N}(\mathbf{0}, \boldsymbol{R}_{\mathrm{s}}[t]), \tag{39}$$

with the quality-aware approximation

$$\boldsymbol{R}_{\mathrm{s}}[t] = \frac{1}{\max\{\gamma_{\mathrm{s}}[t], \gamma_0\}}\,\mathrm{diag}(c_r, c_\vartheta, c_\nu). \tag{40}$$

An extended Kalman filter (EKF), unscented Kalman filter, or particle filter can be used. For the EKF,

$$\widehat{\boldsymbol{x}}_{t|t-1} = \boldsymbol{F}\widehat{\boldsymbol{x}}_{t-1|t-1}, \qquad \boldsymbol{P}_{t|t-1} = \boldsymbol{F}\boldsymbol{P}_{t-1|t-1}\boldsymbol{F}^{\mathrm{T}} + \boldsymbol{Q}_{\mathrm{T}}, \tag{41}$$

$$\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\boldsymbol{H}^{\mathrm{T}}_t(\boldsymbol{H}_t\boldsymbol{P}_{t|t-1}\boldsymbol{H}^{\mathrm{T}}_t + \boldsymbol{R}_{\mathrm{s}}[t])^{-1}, \tag{42}$$

$$\widehat{\boldsymbol{x}}_{t|t} = \widehat{\boldsymbol{x}}_{t|t-1} + d_{\mathrm{s}}[t]\boldsymbol{K}_t\big(\boldsymbol{z}_{\mathrm{s}}[t] - \boldsymbol{h}(\widehat{\boldsymbol{x}}_{t|t-1})\big), \tag{43}$$

$$\boldsymbol{P}_{t|t} = (\boldsymbol{I} - d_{\mathrm{s}}[t]\boldsymbol{K}_t\boldsymbol{H}_t)\boldsymbol{P}_{t|t-1}, \tag{44}$$

where $\boldsymbol{H}_t$ is the Jacobian of $\boldsymbol{h}$. If detection fails, the equations reduce to prediction only.

6.3 On-board computation and delayed availability

All AoA/target-state processing is performed on the HAPS; no MEC station is required. This is realistic only if its bounded compute and processing delay are modelled. Let $C_{\mathrm{s}}[t]$ be the required CPU cycles and $f_{\mathrm{H}}[t] \in [f_{\mathrm{H}}^{\min}, f_{\mathrm{H}}^{\max}]$ the allocated clock frequency. Then

$$d_{\mathrm{p}}[t] = \max\left\{1, \left\lceil \frac{C_{\mathrm{s}}[t]}{f_{\mathrm{H}}[t]T_{\mathrm{s}}} \right\rceil\right\}, \qquad E_{\mathrm{H}}^{\mathrm{cmp}}[t] = \kappa_{\mathrm{H}} C_{\mathrm{s}}[t] f_{\mathrm{H}}^2[t]. \tag{45}$$

Hence a measurement timestamped in slot τ becomes available no earlier than slot $\tau + d_{\mathrm{p}}[\tau]$. It is accepted only if it was detected and its posterior uncertainty passes a quality gate, e.g., $\mathrm{tr}(\boldsymbol{P}_{\tau|\tau}) \le P_{\mathrm{acc}}$. If $g_{\mathrm{s}}[t]$ is the newest accepted measurement timestamp available at the start of slot t, its state is predicted forward to t:

$$\widehat{\boldsymbol{x}}_{\mathrm{T}}^{\mathrm{av}}[t] = \boldsymbol{F}^{t-g_{\mathrm{s}}[t]} \widehat{\boldsymbol{x}}_{g_{\mathrm{s}}|g_{\mathrm{s}}}, \tag{46}$$

$$\boldsymbol{P}^{\mathrm{av}}[t] = \boldsymbol{F}^{t-g_{\mathrm{s}}} \boldsymbol{P}_{g_{\mathrm{s}}|g_{\mathrm{s}}} (\boldsymbol{F}^{t-g_{\mathrm{s}}})^{\mathrm{T}} + \sum_{i=0}^{t-g_{\mathrm{s}}-1} \boldsymbol{F}^i \boldsymbol{Q}_{\mathrm{T}} (\boldsymbol{F}^i)^{\mathrm{T}}. \tag{47}$$

This available belief, not the unprocessed echo, drives the AAV and beam design.

7 AoI, AoLI, and AoSI dynamics

All ages are slot-based and capped for numerical stability. The caps should be set above the operationally meaningful range, not used to hide instability.

7.1 Legitimate age of information

Let $\Delta_{m,k}^{\mathrm{I}}[t]$ be the age at device (m,k) of its most recently delivered status update. With generate-at-will one-slot packets,

$$\Delta_{m,k}^{\mathrm{I}}[t+1] = \begin{cases} 1, & \chi_{m,k}[t] = 1, \\ \min\{\Delta_{m,k}^{\mathrm{I}}[t] + 1, \Delta_{\mathrm{I}}^{\max}\}, & \text{otherwise.} \end{cases} \tag{48}$$

For queued packets, the reset value 1 is replaced by the delivered packet's system time $t + 1 - g_{m,k}[t]$.

7.2 Age of leaked information

Let $\Delta_{m,k}^{\mathrm{L}}[t]$ be the age, at the target/eavesdropper, of the freshest update it has intercepted from device (m,k). Then

$$\Delta_{m,k}^{\mathrm{L}}[t+1] = \begin{cases} 1, & \ell_{m,k}[t] = 1, \\ \min\{\Delta_{m,k}^{\mathrm{L}}[t] + 1, \Delta_{\mathrm{L}}^{\max}\}, & \text{otherwise.} \end{cases} \tag{49}$$

A larger AoLI is desirable because any leaked knowledge is older. AoLI complements, but does not replace, physical-layer secrecy constraints.

7.3 Age of sensing information

Here *AoSI* is defined precisely as the time elapsed since the target state underlying the newest *accepted and available* estimate was sensed. Let

$$\mathcal{C}(t) = \left\{\tau : \tau + d_{\mathrm{p}}[\tau] = t,\ d_{\mathrm{s}}[\tau] = 1,\ \mathrm{tr}(\boldsymbol{P}_{\tau|\tau}) \le P_{\mathrm{acc}}\right\}. \tag{50}$$

To prevent a slowly processed old job from replacing a fresher belief, define

$$\mathcal{C}^+(t) = \{\tau \in \mathcal{C}(t) : \tau > g_{\mathrm{s}}[t]\}. \tag{51}$$

If $\mathcal{C}^+(t) \neq \emptyset$, set $g_{\mathrm{s}}^+[t] = \max \mathcal{C}^+(t)$; otherwise there is no newly useful estimate. The sensing age evolves as

$$\Delta^{\mathrm{S}}[t+1] = \begin{cases} t + 1 - g_{\mathrm{s}}^+[t], & \mathcal{C}^+(t) \neq \emptyset, \\ \min\{\Delta^{\mathrm{S}}[t] + 1, \Delta_{\mathrm{S}}^{\max}\}, & \text{otherwise.} \end{cases} \tag{52}$$

Thus, AoSI includes sensing, detection, and HAPS processing delay. It must be retained together with $\mathrm{tr}(\boldsymbol{P}^{\mathrm{av}}[t])$: two estimates can have the same timestamp age but different accuracy, and the same covariance trace but different ages/prediction risks.

8 AAV mobility, energy, and safe operation

The AAV follows

$$\boldsymbol{q}_{\mathrm{J}}[t+1] = \boldsymbol{q}_{\mathrm{J}}[t] + T_{\mathrm{s}}\boldsymbol{v}_{\mathrm{J}}[t], \tag{53}$$

$$\|\boldsymbol{v}_{\mathrm{J}}[t]\| \le V_{\mathrm{J}}^{\max}, \qquad \|\boldsymbol{v}_{\mathrm{J}}[t+1] - \boldsymbol{v}_{\mathrm{J}}[t]\| \le a_{\mathrm{J}}^{\max} T_{\mathrm{s}}, \tag{54}$$

$$\boldsymbol{q}_{\mathrm{J}}[t] \in \mathcal{Q}_{\mathrm{J}}, \qquad \|\boldsymbol{q}_{\mathrm{J}}[t] - \boldsymbol{q}_{m,k}\| \ge d_{\mathrm{safe}}. \tag{55}$$

For a rotary-wing AAV, a widely used propulsion model is

$$P_{\mathrm{prop}}(v) = P_0\left(1 + \frac{3v^2}{U_{\mathrm{tip}}^2}\right) + P_i\left(\sqrt{1 + \frac{v^4}{4v_0^4}} - \frac{v^2}{2v_0^2}\right)^{1/2} + \frac{1}{2}d_0\rho s A v^3. \tag{56}$$

The battery state is

$$E_{\mathrm{J}}[t+1] = E_{\mathrm{J}}[t] - T_{\mathrm{s}}\left(P_{\mathrm{prop}}(\|\boldsymbol{v}_{\mathrm{J}}[t]\|) + \frac{p_{\mathrm{J}}[t]}{\eta_{\mathrm{PA}}} + P_{\mathrm{c,J}}\right), \quad E_{\mathrm{J}}[t] \ge E_{\mathrm{reserve}}. \tag{57}$$

The AAV may update its velocity only every T_{J} slots, whereas radio power and beam direction can change every slot. This implements a realistic slow mobility/fast radio hierarchy without selecting a learning algorithm.

9 Causal state, action, and policy

At the start of slot t, the controller information is

$$\mathcal{I}_t = \sigma\big(\{\widehat{\boldsymbol{g}}_v[t], \boldsymbol{C}_v[t]\}_v, \{\Delta_{m,k}^{\mathrm{I}}[t], \Delta_{m,k}^{\mathrm{L}}[t]\}_{m,k}, \Delta^{\mathrm{S}}[t], \widehat{\boldsymbol{x}}_{\mathrm{T}}^{\mathrm{av}}[t], \boldsymbol{P}^{\mathrm{av}}[t],$$
$$\boldsymbol{q}_{\mathrm{J}}[t], \boldsymbol{v}_{\mathrm{J}}[t], E_{\mathrm{J}}[t], \boldsymbol{\Theta}[t-1], \text{acknowledgements and completed sensing jobs up to } t\big). \tag{58}$$

The control action is

$$\boldsymbol{a}[t] = \big\{\{a_m, \boldsymbol{v}_m, p_{m,\mathrm{n}}, p_{m,\mathrm{f}}\}_{m=1}^{M}, \boldsymbol{w}_{\mathrm{s}}, \boldsymbol{u}_{\mathrm{s}}, \boldsymbol{\Theta}, p_{\mathrm{J}}, \boldsymbol{u}_{\mathrm{J}}, \boldsymbol{v}_{\mathrm{J}}, f_{\mathrm{H}}\big\}[t]. \tag{59}$$

A feasible policy π is a sequence of mappings $\boldsymbol{a}[t] = \pi_t(\mathcal{I}_t)$; hence every action is $\mathcal{I}_t$-measurable. This is the formal causality constraint.

10 Long-term optimisation problem

10.1 Normalised performance measures

For a process $X[t]$, define $\overline{X} \triangleq \limsup_{T\to\infty} \frac{1}{T}\mathbb{E}_\pi[\sum_{t=0}^{T-1} X[t]]$. Use importance weights $\alpha_{m,k} \ge 0$ with $\sum_{m,k}\alpha_{m,k} = 1$ and define

$$\widetilde{\Delta}^{\mathrm{I}}[t] = \sum_{m,k}\alpha_{m,k}\frac{\Delta_{m,k}^{\mathrm{I}}[t]}{\Delta_{\mathrm{I}}^{\max}}, \tag{60}$$

$$\widetilde{\Delta}^{\mathrm{S}}[t] = \frac{\Delta^{\mathrm{S}}[t]}{\Delta_{\mathrm{S}}^{\max}}, \qquad \widetilde{U}[t] = \frac{\mathrm{tr}(\boldsymbol{P}^{\mathrm{av}}[t])}{P_{\mathrm{ref}}}, \tag{61}$$

$$E_{\mathrm{sys}}[t] = T_{\mathrm{s}}\,\mathrm{tr}(\boldsymbol{Q}_{\mathrm{H}}[t]) + E_{\mathrm{H}}^{\mathrm{cmp}}[t] + T_{\mathrm{s}}\left(P_{\mathrm{prop}}(\|\boldsymbol{v}_{\mathrm{J}}[t]\|) + \frac{p_{\mathrm{J}}[t]}{\eta_{\mathrm{PA}}} + P_{\mathrm{c,J}}\right), \tag{62}$$

$$\widetilde{E}[t] = E_{\mathrm{sys}}[t]/E_{\mathrm{ref}}. \tag{63}$$

The primary cost is

$$J(\pi) = \omega_{\mathrm{I}}\overline{\widetilde{\Delta}^{\mathrm{I}}} + \omega_{\mathrm{S}}\overline{\widetilde{\Delta}^{\mathrm{S}}} + \omega_{\mathrm{U}}\overline{\widetilde{U}} + \omega_{\mathrm{E}}\overline{\widetilde{E}}, \quad \sum_i \omega_i = 1, \quad \omega_i \ge 0. \tag{64}$$

Security is imposed principally through explicit AoLI and secrecy-outage constraints rather than by subtracting AoLI inside the objective. This ϵ-constraint form makes the required security level transparent.

10.2 Problem statement

Let $\Gamma_{m,\mathrm{f}}^{\mathrm{SIC}}$ be the required far-stream SIC threshold, $R_{m,k}^{\mathrm{sec,min}}$ a secrecy-rate target, and $\epsilon_{m,k}^{\mathrm{sec}}$, ϵ_{s} permissible long-term outage levels. The causal stochastic control problem is

$$
\begin{aligned}
\text{(P1)} \quad & \underset{\pi}{\text{minimise}} \quad && J(\pi) && && \text{(65a)}\\
& \text{subject to} && a_m[t] \in \{0,1\}, \quad \sum_m a_m[t] \le 1, && \forall t, && \text{(65b)}\\
& && p_{m,\mathrm{f}}[t] \ge p_{m,\mathrm{n}}[t] \ge 0, && \forall m,t, && \text{(65c)}\\
& && a_m[t]\gamma_{m,\mathrm{n}\to\mathrm{f}}[t] \ge a_m[t]\Gamma_{m,\mathrm{f}}^{\mathrm{SIC}}, && \forall m,t, && \text{(65d)}\\
& && \mathrm{tr}(\boldsymbol{Q}_{\mathrm{H}}[t]) \le P_{\mathrm{H}}^{\max}, \quad 0 \le p_{\mathrm{J}}[t] \le P_{\mathrm{J}}^{\max}, && \forall t, && \text{(65e)}\\
& && \|\boldsymbol{v}_m[t]\| = 1, \quad \|\boldsymbol{u}_{\mathrm{s}}[t]\| = 1, \quad \|\boldsymbol{u}_{\mathrm{J}}[t]\| = 1, && \forall m,t, && \text{(65f)}\\
& && \theta_\ell[t] \in \mathcal{F}_{b_\theta}, \quad T_{\mathrm{d}}[t] \ge T_{\min}, && \forall \ell,t, && \text{(65g)}\\
& && \overline{a_m \mathbf{1}\{R_{m,k}^{\mathrm{sec}} < R_{m,k}^{\mathrm{sec,min}}\}} \le \epsilon_{m,k}^{\mathrm{sec}}\, \overline{a_m}, && \forall m,k, && \text{(65h)}\\
& && \overline{\Delta_{m,k}^{\mathrm{L}}} \ge \Delta_{m,k}^{\mathrm{L,min}}, && \forall m,k, && \text{(65i)}\\
& && \overline{\chi_{m,k}} \ge \chi_{m,k}^{\min}, && \forall m,k, && \text{(65j)}\\
& && \limsup_{T\to\infty} \frac{1}{T}\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \mathbf{1}\{\gamma_{\mathrm{s}}[t] < \Gamma_{\mathrm{s}}\}\right] \le \epsilon_{\mathrm{s}}, && && \text{(65k)}\\
& && \overline{\mathrm{tr}(\boldsymbol{P}^{\mathrm{av}})} \le P^{\max}, && && \text{(65l)}\\
& && f_{\mathrm{H}}^{\min} \le f_{\mathrm{H}}[t] \le f_{\mathrm{H}}^{\max}, && \forall t, && \text{(65m)}\\
& && (53),\ (55),\ (57), && \forall t, && \text{(65n)}\\
& && (48),\ (49),\ (52), && \forall m,k,t, && \text{(65o)}\\
& && \boldsymbol{a}[t] \text{ is } \mathcal{I}_t\text{-measurable.} && \forall t. && \text{(65p)}
\end{aligned}
$$

If the target is silent during some intervals, constraint (65i) remains meaningful because it requires its accumulated knowledge to stay stale. If a hard instantaneous secrecy guarantee is required, replace (65h) by $R_{m,k}^{\mathrm{sec}}[t] \ge R_{m,k}^{\mathrm{sec,min}}$ whenever $a_m[t] = 1$; the outage form is more realistic under uncertain NTN channels.

10.3 Why the problem is difficult

Problem (P1) is a mixed-integer, non-convex, partially observed, long-horizon stochastic control problem. Its difficulty follows from:

- binary pair scheduling and discrete RIS phases;
- non-convex NOMA, secrecy, sensing, and unit-norm beam constraints;
- multiplicative coupling among $\boldsymbol{\Theta}$, HAPS beams, NOMA powers, and AAV jamming;
- delayed target observations and belief-state uncertainty;
- recursive AoI/AoLI/AoSI states whose present actions affect future cost;
- fast radio variables, intermediate RIS/compute variables, and slow AAV motion;
- imperfect CSI, residual SI, and random sensing failures.

Accordingly, the natural exact formulation is a constrained belief-state partially observable Markov decision process (POMDP); if the available belief $\{\widehat{\boldsymbol{x}}_{\mathrm{T}}^{\mathrm{av}}, \boldsymbol{P}^{\mathrm{av}}\}$ and channel estimates are treated as a sufficient observed state, it becomes a constrained MDP. Small instances can be benchmarked with exhaustive/discretised dynamic programming, alternating optimisation, stochastic successive convex approximation, or Lyapunov/queueing methods. A scalable hierarchical learning method is a later design choice, not an assumption embedded in the model.

11 Recommended baseline and progressive extensions

For the first publishable implementation, use perfect knowledge of fixed UE positions, one target, fixed NOMA pairs, a fixed RIS position, Rician channels with imperfect instantaneous CSI, EKF tracking, finite RIS phases, and the complete causal age dynamics above. Retain residual SI, imperfect SIC, AAV energy, and HAPS processing delay because they materially affect conclusions.

The following should be introduced only after validating that baseline:

1. dynamic NOMA pairing and multi-pair spatial multiplexing;
2. several targets/eavesdroppers and data association;
3. movable RIS placement and RIS propulsion energy;
4. explicit finite-blocklength transmission and semantic-value-weighted updates;

5. MEC/cloud offloading or distributed HAPS cooperation;
6. hardware-in-the-loop validation and measurement-calibrated echo/SI models.

12 Core notation

Symbol	Meaning
a_m, $\mathcal{U}_m$	scheduled-pair indicator and fixed near–far NOMA pair
$\boldsymbol{\Theta}$, $\mathcal{F}_{b_\theta}$	aerial-RIS coefficient matrix and finite phase alphabet
$\boldsymbol{v}_m$, $p_{m,k}$	pair communication beam and NOMA powers
$\boldsymbol{w}_{\mathrm{s}}$, $\boldsymbol{u}_{\mathrm{s}}$	sensing transmit beam and HAPS echo combiner
p_{J}, $\boldsymbol{u}_{\mathrm{J}}$	AAV jamming power and beam
ρ_{SI}, ϵ_{SIC}	residual HAPS SI factor and residual NOMA SIC fraction
$\chi_{m,k}$, $\ell_{m,k}$	legitimate delivery and target-interception indicators
Δ^{I}, Δ^{L}, Δ^{S}	AoI, AoLI, and AoSI
$\widehat{\boldsymbol{x}}_{\mathrm{T}}^{\mathrm{av}}$, $\boldsymbol{P}^{\mathrm{av}}$	latest available target-state estimate and covariance
d_{p}, f_{H}	HAPS processing delay and CPU frequency
$\mathcal{I}_t$, π	causal information set and control policy

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